sEMG Muscle Activation Report

Participant P4 · Gesture_1 · April 17, 2026

Overall Activation	Very High
Temporal Profile	Ramp-up
Consistency	Moderate
Dominant Sensors	S4, S8
Weakest Sensors	S5, S6
Assessment Confidence	0.9

Clinical Summary: The session exhibits a clear ramp-up activation profile with high co-contraction, suggesting a power-grip or high-effort task. Motor unit recruitment indicators are elevated in S8 and S4. The high inter-channel stability suggests a consistent, albeit fatiguing, neuromuscular strategy.

Key Findings:
- High co-contraction ratio (>0.5)
- Significant ramp-up profile in S4 and S8
- High variability in S3

Session Overview

Session Duration	1202.4 s (20.0 min)
Sensors	8 (S1–S4 flexor, S5–S8 extensor)
Sampling Rate	1259.3 Hz
Session Mean RMS	0.019717 mV
Feature Source	raw_signal

Per-Sensor Activation Analysis

Muscle activity was recorded across 8 sensors. Sensors S1–S4 are positioned on the palmar/flexor compartment and S5–S8 on the dorsal/extensor compartment. RMS (Root Mean Square) reflects overall activation intensity; MNF and MDF (Mean and Median Frequency) characterise the spectral content of the EMG signal; WAMP (Willison Amplitude) indicates motor-unit firing activity.

Sensor	Mean RMS (mV)	CV	MNF (Hz)	MDF (Hz)	WAMP	Level	Profile
S1	0.018533	0.374	106.8	91.8	1538	High	Sustained
S2	0.017613	0.275	134.2	125.4	3035	High	Sustained
S3	0.023059	0.663	57.8	50.0	1662	Very High	Intermittent
S4	0.031965	0.286	56.9	50.3	2651	Very High	Ramp-up
S5	0.011992	0.186	119.3	99.8	1421	Moderate	Sustained
S6	0.013168	0.381	108.8	89.4	1854	Moderate	Ramp-up
S7	0.015352	0.217	93.4	75.5	1571	High	Sustained
S8	0.026058	0.260	96.4	86.6	4876	Very High	Ramp-up

Flexor sensors (S1–S4) shown with blue labels; extensor sensors (S5–S8) with pink.

Notable Findings per Sensor

S3: High variability in activation

S4: Primary dominant sensor

S8: High WAMP indicates significant recruitment

Spatial Activation Patterns

Cross-channel metrics characterise how muscle activation is distributed across the forearm during the recorded gesture. A **high concentration index** indicates focal activation (one or two dominant sensors); a **low value** suggests diffuse activation. The **flexor/extensor balance** reveals which compartment drives the gesture. **Pattern stability** near 1.0 indicates the spatial pattern is maintained throughout the session, with no substantial load redistribution.

Metric	Value	Interpretation
Dominant Sensor	S4	Primary contributor
Weakest Sensor	S5	Lowest activity
Dominant-to-Weakest Ratio	2.67×	Activation disparity
Concentration Index	0.149	Focal (near 1) vs Diffuse (near 0.125)
Pattern Stability	0.993	Spatial consistency (1 = stable)
Inter-Channel Correlation	0.212	Crosstalk check (>0.8 = concern)
Co-Contraction Ratio	0.747	Agonist/antagonist balance

Flexor/Extensor Balance	1.36	>1 = flexor-dominant
Overall Temporal Profile	Ramp-up	Session-wide activation trend

Co-contraction: Detected — Agonist and antagonist muscles activating simultaneously (normal for power grip).

Spatial Compensation: Not detected — Stable spatial activation pattern throughout the session.

Gesture-Pattern Interpretation: Flexor-dominant activation (1.362 ratio). S4 (flexor) and S8 (extensor) show significant ramp-up profiles, suggesting increasing effort over the session duration.

Motor-Unit Recruitment

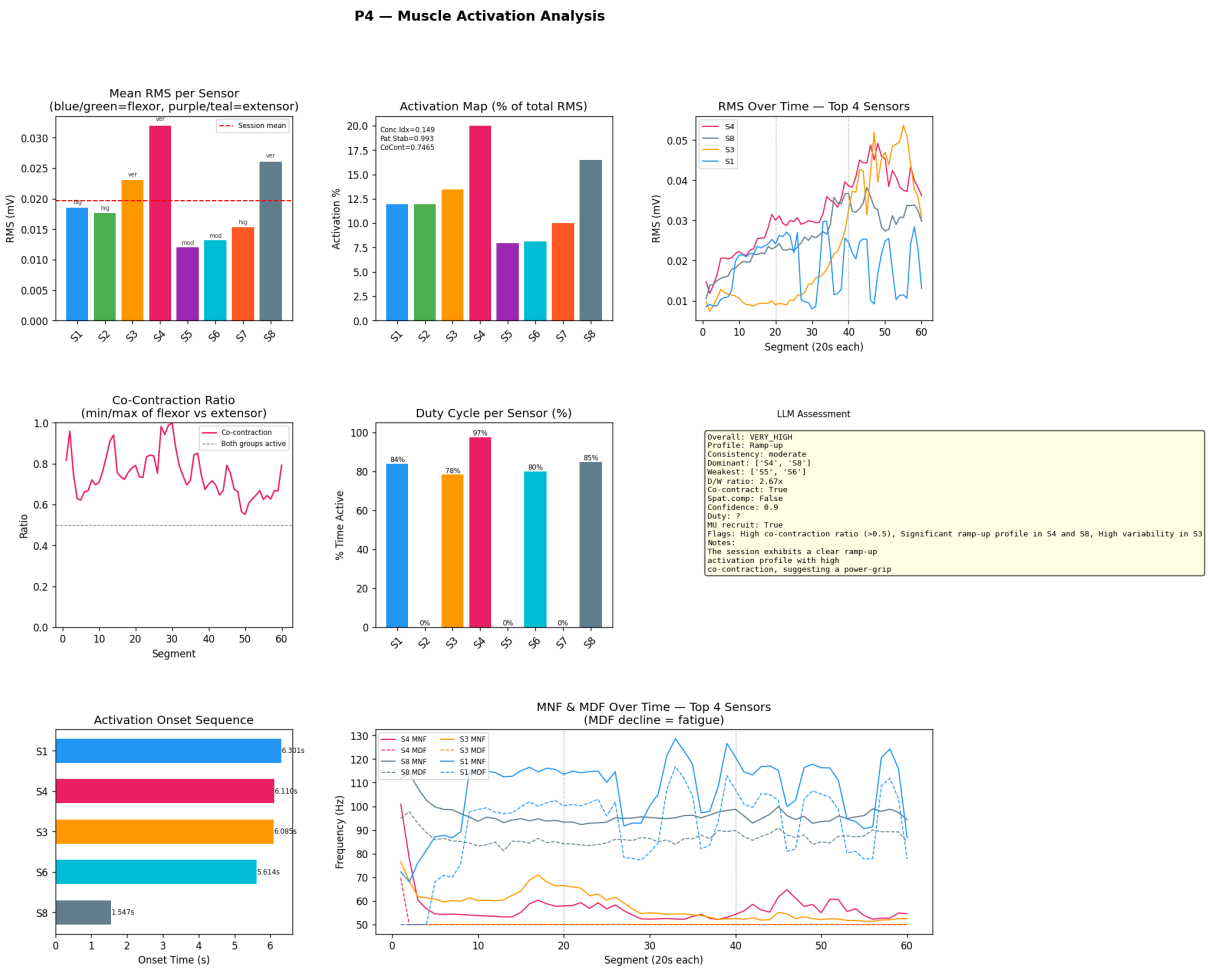
Motor-unit recruitment is inferred from the Willison Amplitude (WAMP) and zero-crossing rate. Elevated WAMP indicates frequent changes in signal amplitude, consistent with the firing of additional (often higher-threshold) motor units as effort or fatigue increases.

Recruitment Detected	Yes
Sensors Affected	S8, S2, S4

Evidence: High WAMP values observed: S8 (4875.6), S2 (3035.1), and S4 (2651.3) suggest increased motor unit recruitment consistent with high-effort or fatigue-related compensation.

Visualisation

The figure below consolidates the session analysis: per-sensor RMS amplitudes, spatial activation map, RMS time series for dominant sensors, co-contraction ratio, and frequency-content (MNF/MDF) trends over the 20-minute session.



Methodology & Limitations

Signal Processing

Raw surface EMG signals from 8 Delsys Avanti sensors were band-pass filtered (20–450 Hz, Butterworth 4th order) to attenuate motion artefacts and high-frequency noise. Twelve features were extracted per sensor in consecutive 20-second windows: RMS, MAV, IEMG, Waveform Length, Zero Crossings, Slope Sign Changes, Willison Amplitude, Peak Amplitude, Variance, Mean Frequency, Median Frequency, and Total Power. Cross-channel metrics (activation map, concentration index, pattern stability, co-contraction ratio, flexor/extensor balance) were computed from the per-sensor RMS time series.

Clinical Interpretation

Feature summaries were passed to a large language model (gemini-3.1-flash-lite-preview) along with a structured set of clinical rules (activation level thresholds, gesture patterns, co-contraction criteria, motor-unit recruitment indicators) and retrieval-augmented research context drawn from a knowledge base of sEMG literature (Konrad 2005, Phinyomark 2012, Vigotsky 2018, Nazmi 2016, Yang 2024 EMGBench).

Limitations

• **No %MVC normalisation:** Amplitudes are absolute (mV); direct cross-subject comparisons of amplitude are unreliable without a reference contraction.
• **Sustained-contraction data:** Because the gesture is held continuously, onset and duty-cycle metrics are not reported.
• **No ground-truth validation:** Findings are descriptive. Clinical decisions should consider additional context (force measurement, movement observation, patient history).
• **LLM output:** Provided as a summarisation aid, not a diagnostic tool. Human-in-the-loop review is required.

Report generated automatically by the sEMG Interpretation Pipeline. For clinical use, please corroborate findings with additional assessments.